Drug Inventory and Supply Chain Tracking System using Blockchain

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***Abstract*—** The system records all drug-related transactions such as manufacturing, distribution, and dispensing in a secure, append-only ledger, where each entry is cryptographically linked to its predecessor to ensure immutability. The inventory details are kept in a detailed stock record in an SQL Server/MySQL database (off-chain storage), and the SHA-256 digests are stored in the ledger-style blockchain to provide the tamper-evidence and traceability. We implement the business logic and APIs in Java so that the application is portable and scalable. The hybrid architecture design helps to increase the trust, enhances the transparency between differenct stakeholders, and enables the instant drug authenticity verification, which can reduce the counterfeit drugs, ensure compliance with regulatory requirements.

***Keywords*— Blockchain, Pharmaceutical Supply Chain, SHA-256, Drug Traceability, Java, Hybrid Architecture**

I. Introduction

Pharmaceutical industry is very sensitive, high regulated and a life-threatening industry as it affects human health and life directly, so the authenticity, availability and quality of medicines at each link of the supply chain is a very complex problem. There are many entities involved from manufacturing to distributing, pharmacies, hospitals, and regulatory bodies to form a complex system, and with globalization, complex distribution channels, and increasing demand, the risks of fake medicines, stock keeping issues, and fraudulent activities have grown exponentially. Counterfeit, fake and substandard drugs as per World Health Organization (WHO), are a major portion of fake medicines, which cause serious side effects and death and a lack of confidence in the healthcare system. Traditional supply chain management databases are very efficient in operational manner but they are vulnerable to tampering, lack of transparency and end-to-end traceability, because a system with a single point of failure and insider malicious can take down the entire system. The use of Blockchain technology can be a game changer, with an immutable distributed ledger, which securely and transparently records every transaction using cryptographic links between the blocks. However, full-scale blockchain implementations may face challenges in scalability, storage, and integration with existing systems. To address these issues, this project proposes a Drug Inventory and Supply Chain Tracking System using Blockchain with SHA-256 hashing, combining blockchain-inspired ledger maintenance with off-chain storage in SQL Server or MySQL for scalability. Implemented in Java, the system leverages robust APIs, cross-platform portability, and enterprise-grade security, allowing detailed records to be efficiently stored in a relational database while their integrity is ensured through cryptographic hashing and linked ledger 1 entries. This hybrid approach enhances transparency, accountability, and trust among stakeholders while offering a practical, cost-effective solution for real-world adoption in the pharmaceutical supply chain. However, implementing a full-scale public or permissioned blockchain network for pharmaceutical inventory faces several practical challenges: **scalability issues** (due to the large volume of transactions), **high data storage costs** (since every node stores the full ledger), and **integration complexity** with existing enterprise IT infrastructure that relies heavily on fast relational databases.

II. EXISTING SYSTEM

Current drug inventory and supply chain management are heavily reliant on centralized databases, manual record keeping and barcode-based tracking methods.

While these methods provide basic inventory control and transaction recording functions, their transparency is low, are susceptible to manipulation, and supply chain traceability is limited at best. Centralized systems, maintained and controlled by a single party, are vulnerable to fraud, manipulation and even system failure. As a result, counterfeit drugs often infiltrate the market due to the absence of comprehensive mechanisms for end-to-end verification of drug authenticity. Manual processes further contribute to inefficiencies, delays in product recalls, and inadequate monitoring of critical cold-chain requirements such as temperature and storage conditions. While some organizations have adopted ERP or cloud-based platforms for inventory management, these solutions still fall short of providing tamper-proof audit trails and seamless interoperability among various stakeholders. Consequently, ensuring trust, security, and regulatory compliance in pharmaceutical distribution remains a significant challenge.

III. PROPOSED SYSTEM

The proposed system introduces a hybrid drug inventory and supply chain tracking framework that integrates blockchain inspired ledger maintenance with the SHA-256 hashing algorithm to ensure security, transparency, and tamper-evidence. Unlike the traditional centralized systems where all the drug transaction information (batch details, manufacturing and expiry dates, stock, and distribution information) is stored in a relational database (SQL Server/MySQL) for persistence and scalability. On the other hand, every transaction is hashed with SHA-256 to create a digital fingerprint that is stored in an append-only ledger table along with the timestamp, actor ID, and previous hash (like a chain). The entire system is developed using Java, which provides robust APIs, secure transaction handling, and easy integration with the backend database. By integrating off-chain database storage and on chain hash validation, the system ensures that when an attacker tries to modify the database records, the modification will be caught and detected during the hash validation process. In addition, the system also implements role-based access control for various roles, including manufacturers, distributors, pharmacies, and regulators, to achieve end-to-end traceability and regulatory compliance. Finally, the proposed system design offers an integrated solution that is practical, affordable, secure, and can improve trust, anti-counterfeit drugs, and accountability in pharmaceutical supply chain.

IV. MODULES OF THE PROJECT

The source code for Drug Inventory and Supply Chain Tracking System using Blockchain with SHA-256 consists of multiple modules. The modules are very important in maintaining the security, integrity, and efficiency of the supply chain.

1. User Authentication and Access Control Module

Handles the registration and login of entities such as manufacturers, distributors, pharmacies, and regulators.
Handles role-based access control based on Java security and Spring Security. Ensures that only authorized users can perform specific operations (e.g., manufacturers can create batches, regulators can audit records).

2. Drug Batch Management Module

Allows manufacturers to create and register new drug batches with details such as batch number, drug name, manufacturing date, expiry date, and quantity.
Stores all batch information in the SQL Server/MySQL database.
Generates a unique identifier for each drug batch to support traceability

3. Transaction and Inventory Management Module

Manages supply chain transactions such as manufacturing, stock transfer, distribution, and sales at the pharmacy level.
Updates inventory levels at each stakeholder's end.
Maintains history of movement for each drug batch across the supply chain.

4. Ledger Maintenance Module

Implements a blockchain-inspired append-only ledger.
For each transaction, generates a SHA-256 hash and stores it with metadata (transaction ID, timestamp, actor ID, previous hash) in the ledger table.
Ensures immutability by linking each ledger entry to the previous one.

5. Hashing and Verification Module

Computes SHA-256 hash for each transaction record.
Provides verification functionality by recalculating hashes and comparing them with stored ledger entries.
Detects tampering or unauthorized modifications in database records.

6. Audit and Reporting Module

Allows regulators and administrators to track the entire lifecycle of a drug batch.
Generates reports on stock levels, distribution paths, expiry alerts, and suspicious activity.
Provides visualization of supply chain flow and compliance checks.

7. Notification and Alert Module (Optional/Future Enhancement)

Sends alerts for near-expiry drugs, inventory shortages, or mismatched hash values detected during verification.
Helps stakeholders take timely action to avoid risks.

V. IMPLEMENTATION

The implementation of the Drug Inventory and Supply Chain Tracking System using Blockchain with SHA-256 is carried out in a structured and layered manner. The system is developed using Java as the primary programming language, with SQL Server/MySQL as the backend database, and incorporates blockchain-inspired ledger maintenance to ensure security and immutability of supply chain records.
Frontend layer is written in Java Swing, JSP and HTML with Servlets providing a front end with a platform where the user through an intuitive dashboard can perform actions such as registering of new drug batches, updating transaction, checking stock and audit functions. This

frontend layer is role based to the needs of manufacturers, distributors, pharmacies and regulators.

The backend layer is designed using Spring Boot to handle business logic and application services. The critical feature of the backend is the integration of the SHA-256 hashing algorithm, implemented through the Java Message Digest class. For every transaction performed in the system, a hash value is generated and securely stored, thereby creating a chain of linked records. This mechanism ensures that each transaction is tied to the previous one, forming a blockchain-like structure that preserves data integrity and prevents tampering.

The database layer is built on relational database systems such as SQL Server or MySQL. It maintains tables for drug batches, stakeholders, transaction records, and the blockchain-inspired ledger. Database design includes stored procedures and triggers, which ensure transactional consistency and maintain synchronization between records. This layer serves as the central repository for both operational and security-related data.

A dedicated ledger maintenance and verification module ensures that each new transaction is first entered into the transaction table and subsequently hashed using SHA-256. The generated hash is then stored in the ledger table along with metadata such as timestamp, transaction ID, and previous hash. During verification, the system recalculates the hashes and compares them with stored values, immediately detecting any discrepancies caused by unauthorized modifications.

Finally, security measures are enforced through Spring Security, which provides authentication and role-based authorization. The implementation of TLS/SSL protocols further secures communication between client and server, while the use of SHA-256 hashing ensures that the integrity of all stored data is preserved. Together, these features create a secure and transparent framework for pharmaceutical supply chain management.

VII.PERFORMANCE ANALYTICS

In this paper, the hybrid system, which combines blockchain-inspired ledger maintenance with the SHA-256 hashing algorithm and an off-chain SQL database, achieves high security and reliability without compromising performance.

Key Performance and Efficiency Findings:

Negligible Delay from Hashing:

The core finding is that the cryptographic process the **SHA-256 hashing and ledger maintenance** did not introduce a bottleneck to the system's operation.

Mechanism: Every time a transaction (e.g., drug batch transfer, dispensing) occurs, the system compiles the transaction data, calculates its **SHA-256 hash**, and appends this compact hash along with the link to the previous hash into the ledger.

Result: This entire process is completed so rapidly that the resulting delay is described as **negligible**.

Implication: This low latency ensures the system is **suitable for real-time operations** in environments like pharmacies or distribution centers where immediate inventory updates and transaction recording are critical

Efficient Transaction Handling:

The system was designed to manage the high volume of daily transactions characteristic of a complex supply chain effectively.

Efficiency Metric: The system demonstrates an ability to **process and validate multiple transactions concurrently** without system slowdown or data corruption.

Contrasting with Full Blockchain: Unlike full-scale blockchain systems where every node must validate every transaction, the hybrid model delegates the bulk storage and standard transactional tasks to the **high-speed SQL database**. This separation allows the system to focus on the essential security steps (hashing) efficiently.

Benefit: This efficiency contributes directly to **improved inventory accuracy** because updates occur instantly as products move through the supply chain

Comparison to Traditional Systems:

The project successfully demonstrates that its security enhancements were achieved *without* sacrificing performance relative to standard database solutions.

Security Gain: The blockchain-inspired ledger design and cryptographic linking (SHA-256) provide a significant boost in **security and data reliability** by making records tamper-evident.

Performance Parity: Crucially, the system was found to maintain performance that is **on par with a traditional, non-secured SQL-only database system**.

Value Proposition: The system delivers a **"best of both worlds"** scenario: the speed and familiarity of relational databases combined with the robust, immutable security principles of blockchain.

Scalability:

Scalability is the central advantage of adopting the hybrid architecture, addressing a major drawback of many pure blockchain implementations.

Hybrid Design: The model ensures long-term scalability by making a clear distinction between data storage:

Bulk Drug Records (Off-Chain): Detailed, voluminous records (drug names, expiry dates quantities, etc.) are stored in the highly scalable SQL database (SQL Server/MySQL). This keeps retrieval and reporting fast.

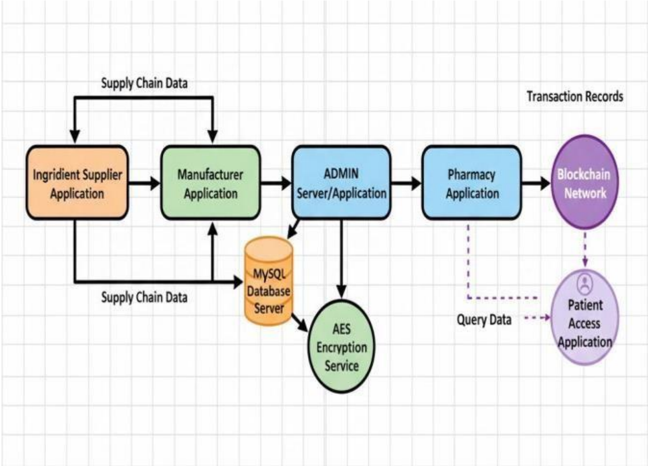
Security Records (Ledger): Only **compact, small-sized hash values** and linking pointers are stored in the secure ledger.

Benefit: As the volume of drug transactions grows exponentially, the bulk of the data (the largest factor for storage and latency) is handled efficiently by the SQL database, preventing the ledger itself from becoming cumbersome, costly, or slow a common issue in traditional blockchain scaling.

Immutability with Tamper Detection:

While a security feature, tamper detection is a critical part of the system's operational integrity and audit performance.

Security Feature: The primary security mechanism is the **cryptographic link** between transactions.



VI. CONCLUSION

The proposed **Drug Inventory and Supply Chain Tracking System using Blockchain with SHA-256** effectively addresses issues of transparency, security, and accountability in pharmaceutical supply chains. By combining blockchain-inspired ledger maintenance with cryptographic hashing, the system ensures all drug transactions are **immutable, traceable, and tamper-proof**.

Implemented using **Java** with **SQL Server/MySQL** as the backend, it offers a reliable and scalable environment. Each transaction is securely linked through **SHA-256 hashes**, making any unauthorized modifications easily detectable and preventing counterfeit records.

The system provides **real-time updates** on drug batches and stock levels across manufacturers, distributors, and pharmacies, improving inventory accuracy and reducing wastage from expired drugs. Its **audit and reporting features** support regulatory oversight and enhance operational efficiency.

Results show that the system not only strengthens supply chain security but also establishes a **practical framework** for real-world implementation. It minimizes counterfeit risks, ensures compliance with regulations, and promotes transparency among all stakeholders. In conclusion, integrating blockchain principles with secure hashing creates a **robust foundation for future enhancements**, such as IoT-based tracking, smart contracts for automated compliance, and cloud deployment for broader accessibility.

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